Fiscal Year:	FY 2025	**Task Last Updated:**	FY 10/09/2024
PI Name:	Greenhalgh, Preston C. M.S.		
Project Title:	HF Occupant Protection		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **Dynamic Loads**:Risk of In-Mission Injury and Performance Decrements and Long-term Health Effects due to Dynamic Loads		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	GOVERNMENT	**Phone:**	281-483-6639
Organization Name:	KBR Wyle/NASA Johnson Space Center		
PI Address 1:	JSC B21		
PI Address 2:	2400 NASA Parkway		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2015	**End Date:**	10/01/2031
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	2024 Update: Preston Greenhalgh is the new Principal Investigator for this Directed Research project. The Institutional Review Board (IRB) protocol was updated in February 2024 to reflect the change. For earlier reporting, see "Soyuz Landing Injury Risk Characterization" under the previous PI, Nathaniel Newby (https://). Nate Newby remains with the project as a Co-Investigator; he is now the Technical Discipline Lead. Rachel Thompson has been added as a Co-Investigator. Kevin Dolick is no longer with the project. (Ed., 10/4/24).		
COI Name (Institution):	Somers, Jeffrey M.S. (KBR/NASA Johnson Space Center) Reiber, Teresa M.S. (KBR/NASA Johnson Space Center) Yates, Keegan Ph.D. (KBR/NASA Johnson Space Center) Putnam, Jacob Ph.D. (NASA Langley Space Center) Newby, Nathaniel M.S. (KBR/NASA Johnson Space Center) Drake, Aaron M. M.S. (KBR/NASA Johnson Space Center) Howes, Caroline E. M.S. (KBR/NASA Johnson Space Center) Thompson, Rachel L. Ph.D. (KBR/NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	October 2024 - Updated Task Description per PI (Ed., 10/11/24) Currently the impact load imparted to crewmembers landing in the Soyuz vehicle is unknown. This study is the first systematic assessment of the number and types of injuries associated with Soyuz landing. To date, we have found that nearly a third of United States Orbital Segment (USOS) crewmembers are experiencing injuries. Most of these injuries are minor, but they exceed expected rates based on analysis of seat accelerometer data from airborne and drop tests of the vehicle. The yet to be answered question is whether spaceflight deconditioning renders crewmembers more susceptible to landing impact injuries. Another possibility is that the Soyuz landing load is higher than our current estimates. It could also be that our analytical tools are insufficient to predict injury rates accurately for space vehicles. A final possibility is that some combination of these factors are responsible. Current occupant protection requirements levied upon future NASA space vehicles are based, in part, on the mathematical prediction that minor injuries due to Soyuz landings have less than a 5% probability of occurrence. Understanding the risk of injury during Soyuz landings and the validity of current crew risk models will allow for improved crew protection in future spacecraft designs. The following are the specific aims for this task: Aim 1: This aim consists of data collection from two sources. One is flight medical records from a database maintained by the NASA Lifetime Surveillance of Astronaut Health (LSAH). This data is only obtainable for US astronauts. The other data source is from a survey that crewmembers are asked to complete. The survey requires an additional consent process. Aim 2: The study team has not had success obtaining Soyuz seat acceleration data, and NASA management has abandoned any future effort of obtaining it. However, NASA is supporting a Small Business Innovation Research (SBIR) grant for the development of miniaturized accelerometers that can be worn by astronauts during dynamic phases of spaceflights. The conceptual design would allow the crew to adhere these sensors to locations such as the head or spine. Aim 3: An analysis has been performed to identify the expected rates of Soyuz injury using the Brinkley model. These expected rates are compared with the actual injury rates uncovered by this study.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Continuation of "HF Occupant Protection" project. As noted, earlier reporting can be found under the previous PI, Nathaniel Newby (Ed., 10/4/24).
Bibliography Type:	Description: (Last Updated: 05/20/2026)
Abstracts for Journals and Proceedings	Newby NJ, Greenhalgh PC, Reiber TM, Somers JT. "Soyuz Landing risk characterization." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024